

Imaging of Primary and Secondary Renal Lymphoma

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OBJECTIVE. This article reviews the CT and MRI patterns of primary and secondary renal lymphomas and discusses the role of percutaneous biopsy in diagnosis and management.

CONCLUSION. Renal lymphoma has a variable imaging spectrum and may mimic renal cell carcinoma. An awareness of the typical and atypical imaging features of both primary and secondary renal lymphomas can help the radiologist to suggest these diagnoses and recommend biopsy when appropriate.

Renal involvement in lymphoma commonly occurs in the presence of widespread nodal or extranodal lymphoma and is classified as secondary renal lymphoma (SRL). However, lymphoma may rarely involve the kidneys alone without evidence of disease elsewhere; then, it is termed “primary renal lymphoma” (PRL) [1, 2]. Although the diagnosis of renal lymphoma can be challenging, an awareness of the spectrum of imaging findings can help to differentiate lymphoma from other renal malignancies such as renal cell carcinoma (RCC) and can lead to appropriate recommendations for biopsy. An accurate diagnosis is critical because renal lymphoma is treated by chemotherapy whereas RCC is typically managed by surgery or ablation.

On ultrasound, renal lymphoma is usually hypoechoic or anechoic [3]. Diffuse nephromegaly may also be seen. However, ultrasound is less sensitive than contrast-enhanced CT for detecting renal lymphoma [3]. Further, sonographic appearances of renal lymphoma are usually nonspecific and lead to further investigation with CT or MRI. This article reviews the CT and MRI patterns of renal lymphoma and discusses the role of percutaneous biopsy in the diagnosis and management.

PRL is exceedingly rare, accounting for fewer than 1% of extranodal lymphomas [2, 4, 5] (Figs. 1 and 2). The fact that the kidneys lack lymphoid tissue has led many to question whether PRL truly exists [1]. Various theories about the origin of PRL have been postulated including an origin of the renal capsule or perinephric fat (which has rich

lymphatics) with subsequent spread to renal parenchyma [6–8]. Chronic inflammation is also reported to be a possible etiologic factor [9, 10]. To avoid misdiagnosing secondary renal involvement as PRL, investigators have reported that PRL should be diagnosed only in the following situations: disease localized to the kidney without evidence of any other organ or nodal involvement; presentation with renal failure in the absence of other causes of renal impairment and rapid improvement of renal function after treatment of lymphoma; or diagnosis confirmed by biopsy [11]. Histopathology shows infiltration of the interstitium with or without glomerular involvement. Pure intraglomerular infiltration may also be present [12]. PRL is reported to occur more commonly in middle-aged men, and patients are usually symptomatic with flank pain, fatigue, or weight loss [13]. Hematuria, proteinuria, and renal failure are often present [14].

In contrast to PRL, SRL commonly occurs in patients with widespread disease. Although autopsy studies report an incidence as high as 30–60%, the imaging manifestations of SRL occur in only 1–8% [15–17]. This discrepancy between autopsy and imaging studies may be explained by a high incidence of clinically silent cases that are diagnosed only after microscopic examination of the kidneys. SRL occurs from either direct or hematogenous spread from disseminated disease. SRL has been reported in both pediatric and adult patients [18].

Although any subtype of lymphoma can involve the kidneys, renal lymphoma (both

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PRL and SRL) is most commonly seen in patients with non-Hodgkin lymphoma, usually the B-cell type or Burkitt lymphoma. Rarely, the kidneys may be involved in Hodgkin lymphoma [19–21]. Immunocompromised status secondary to organ transplantation and HIV infection may predispose patients to renal lymphoma [22, 23].

CT Features

SRL may be unilateral or bilateral and can occur as solitary or multiple focal masses [15, 16, 19, 24–26] (Figs. 3 and 4). SRL may originate from the renal parenchyma or from the perinephric space (Fig. 5). SRL may also present as diffuse bilateral nephromegaly without any focal lesions (Fig. 6). Less frequently, SRL may involve the renal sinus (Fig. 7).

Renal lymphoma is typically a hypovascular tumor. This feature can help to differentiate it from the more common hypervascular tumors such as RCC, oncocytoma, and angiomyolipoma (Fig. 8). MDCT using a renal protocol involving unenhanced, corticomedullary, and nephrographic phase imaging is recommended for diagnosing and characterizing renal lesions, including renal lymphoma [26, 27]. Although the nephrographic phase is probably the most sensitive phase for diagnosing a hypovascular tumor such as renal lymphoma, the multiphasic study helps to exclude other diagnoses. Unenhanced CT can help to differentiate renal lymphoma from a hyperdense benign cyst. A renal mass measuring 70 HU or more on unenhanced images is characteristic of a benign hyperdense cyst, whereas renal lymphomas usually measure 30–50 HU on unenhanced images [15]. Similarly, the corticomedullary phase is useful to differentiate renal lymphoma from clear cell RCC because the latter tends to be hypervascular and is clearly visualized in this phase whereas the hypovascular renal lymphoma may be less conspicuous [26].

Many benign and malignant tumors including oncocytoma, angiomyolipoma, RCC, and transitional cell carcinoma may present as focal renal masses. A history of lymphoma, the hypovascular nature of the mass, and the presence of retroperitoneal adenopathy may help to differentiate SRL from these lesions. Infection (pyelonephritis or abscess) and infarcts may also present as multifocal renal lesions, but clinical correlation is usually helpful. Hematoma, sarcoma, extramedullary hematopoiesis, and metastases are the main differential diagnoses for perinephric masses. Infection, leukemia, and rare renal tumors in-

cluding collecting duct or medullary carcinoma may present with bilateral nephromegaly. In cases of diagnostic dilemma, percutaneous biopsy may be required [28–30].

Unlike SRL, few studies have classified the imaging patterns of PRL because of its relative rarity [25, 31]. Case reports and case series from the urologic and oncologic literature suggest an imaging spectrum similar to that of SRL. PRL may present as focal masses (solitary or multiple); large, ill-defined infiltrative lesions engulfing the kidneys; or diffuse bilateral enlarged kidneys [2, 32–34] (Figs. 1 and 2). PRL occurring purely as a renal sinus mass is extremely rare [35].

PRLs are often large infiltrative renal tumors that extend into the perinephric space and retroperitoneum. However, despite being aggressive tumors, PRLs only rarely involve the inferior vena cava; this characteristic of PRL can help differentiate it from RCC [36] (Fig. 1). Careful evaluation to ensure that the epicenter of the tumor is not within the collecting system is helpful in differentiating PRL from urothelial carcinoma. However, a definitive diagnosis usually requires biopsy [37].

MRI Features

Although CT is the most common imaging modality used for the evaluation of renal lymphoma, MRI may be useful in patients with renal insufficiency or a history of contrast medium allergy or in those in whom there is a heightened concern for radiation exposure such as children and young adults. Renal lymphoma is usually low to intermediate signal intensity on T1- and T2-weighted sequences. However, heterogeneous high signal intensity may be seen on T2-weighted images (Fig. 2). After IV contrast material administration, mild heterogeneous enhancement less than that of the renal cortex is usually seen within the mass [38, 39]. The imaging features of renal lymphoma on diffusion-weighted sequences have not been well established. In our limited experience, renal lymphoma tends to show restricted diffusion (Fig. 2).

Role of PET/CT

PET/CT plays an important role in the evaluation of extranodal lymphoma, particularly in view of its higher sensitivity for diagnosing hepatosplenic involvement [40, 41]. The role of PET/CT in renal lymphoma has not been formally evaluated in any large-scale studies to date [41, 42]. However, PET/CT may be useful in the diagnosis because renal lymphoma is intensely FDG-avid whereas RCC

including the papillary and chromophobe subtypes may not show intense FDG uptake [21] (Figs. 1, 2, and 9). PET/CT is also useful in assessing response to therapy.

Role of Biopsy

Improvements in technique and diagnostic accuracy have resulted in the increased use of percutaneous imaging-guided renal biopsy in the management of incidental renal masses [43–46]. Recent studies indicate a sensitivity and specificity of 90–100% for percutaneous imaging-guided renal biopsy in diagnosing malignancy, which can help avoid unnecessary surgery if a benign diagnosis can be made on biopsy [47–53].

The diagnosis of PRL usually requires biopsy in most cases because imaging findings are not specific and histopathologic analysis drives specific therapies and prognostication. Although fine-needle aspiration is a useful technique, core biopsy with flow cytometry and immunohistochemical staining is recommended because it has a higher diagnostic accuracy [30, 54–56].

With regard to SRL, renal biopsy is usually not required if there is disseminated disease. However, investigators have reported that patients with lymphoma have a higher incidence of RCC than the general population [57–59]. Differentiating SRL presenting as a solitary mass (typically hypovascular) from clear cell RCC (typically hypervascular) may be straightforward. However, other histologic subtypes of RCC such as papillary and chromophobe RCCs may also be hypovascular, making differentiation from SRL difficult. In such cases, a renal biopsy may be justified.

Further, patients with known primary malignancies other than lymphoma may develop synchronous or metachronous lymphoma that may present as a renal mass. When in doubt, renal biopsy may also be useful in those circumstances to differentiate renal lymphoma from renal metastasis because the treatment differs significantly.

Management

Traditionally, PRL was reported to be associated with a poor prognosis [34, 60–62]. However, recent reports suggest that early diagnosis and chemotherapy involving rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisolone (“R-CHOP”) may improve renal function within 2–4 weeks of initiating therapy and may improve 5-year survival rates [63, 64].

In previous studies, investigators have reported that secondary renal involvement in pa-

tients with widespread extranodal lymphoma may not affect prognosis [65]. However, SRL can result in acute renal failure in 6–16% or impaired renal function in nearly one fourth of patients [29, 66]. Morel et al. [29] reported that tumor size larger than 10 cm, involvement of the renal hilum, and diffuse renal infiltration may be associated with a poorer prognosis. Further, recent studies indicate that involvement of the kidneys at the time of initial presentation in diffuse large B-cell lymphoma may be associated with a much higher incidence of CNS relapse, which results in poorer prognosis [66–68]. Hence, it is important to identify renal involvement because patients can then be treated with the addition of rituximab (a monoclonal antibody targeting CD20) to the standard chemotherapy regimen, which may improve both progression-free and overall survival rates [66, 68, 69].

Conclusion

Renal lymphomas are associated with various imaging appearances on CT and MRI. The disease may be unilateral or bilateral and may present as focal masses (solitary or multiple) or diffuse infiltrative lesions or may manifest as enlarged kidneys without focal lesions. Unlike RCC, renal lymphoma is typically hypovascular and does not invade the vessels. Early diagnosis and proper management result in a favorable outcome. Knowledge of the imaging spectrum of PRL and SRL and the potential confounding diagnoses will enable the radiologist to suggest a renal biopsy when necessary, prevent an unnecessary nephrectomy, and also avoid a delay in initiating the appropriate therapy.

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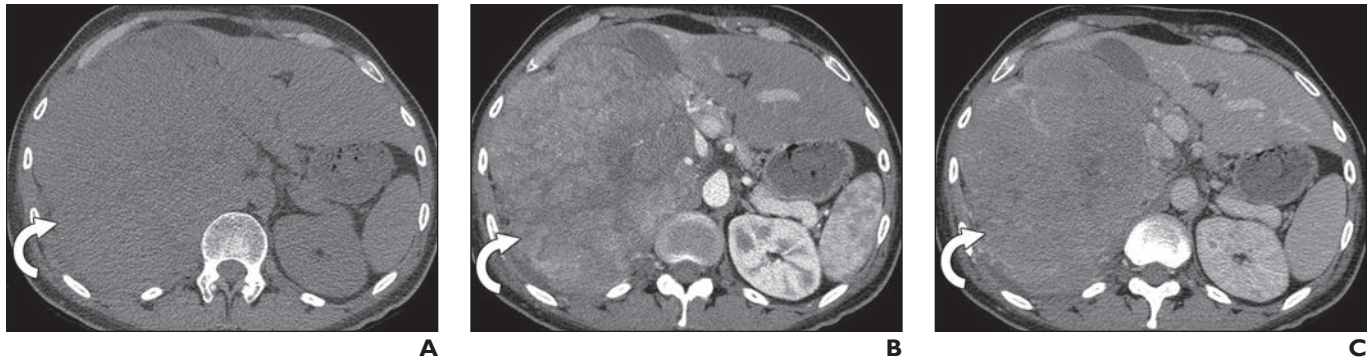


Fig. 1—30-year-old man with HIV infection. **A**, Axial unenhanced CT image shows large mass (arrow) in right retroperitoneum has completely engulfed kidney and is pushing adjacent organs. **B** and **C**, Axial contrast-enhanced CT images in corticomedullary (**B**) and nephrographic (**C**) phases show that mass (arrow) enhances less than adjacent renal cortex. Despite large size of tumor, no invasion of inferior vena cava is seen. **D**, Volume-rendered 3D coronal reformation image shows mass (arrow) is supplied by right renal artery; this finding implies renal origin of mass. **E**, Coronal fused PET/CT image shows lesion (arrow) is markedly FDG-avid. No FDG-avid lesions are seen elsewhere in body. Renal biopsy with immunohistochemical staining and flow cytometry confirmed diagnosis of primary renal lymphoma of plasmablastic subtype.

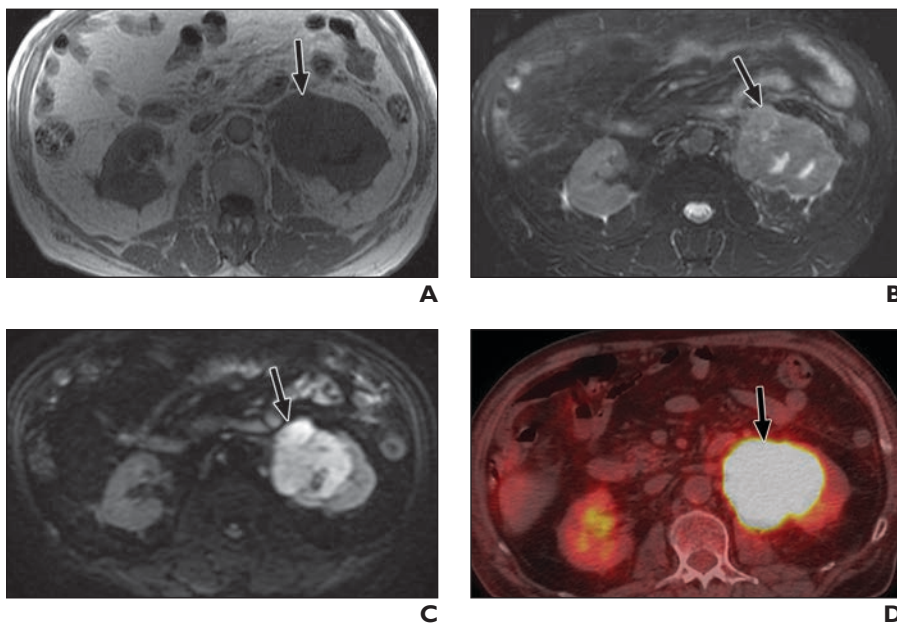
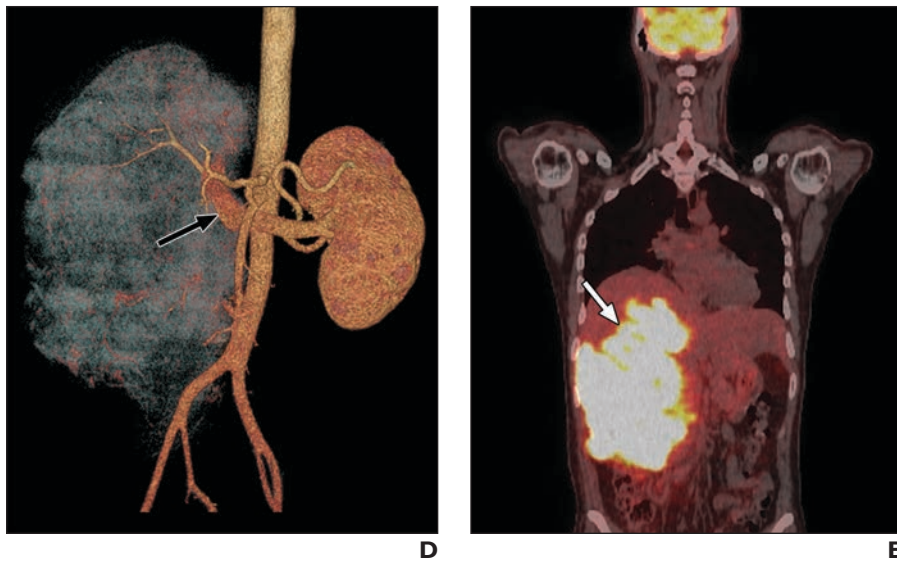


Fig. 2—56-year-old man who presented with left flank pain. **A**, Axial T1-weighted MR image shows 7-cm mass (arrow) in interpolar region of left kidney that is isointense to kidney. **B**, Axial T2-weighted MR image shows heterogeneous high signal intensity in mass (arrow). **C**, Axial diffusion-weighted image shows restricted diffusion in mass (arrow). **D**, Axial fused PET/CT image shows lesion (arrow) is markedly FDG-avid.

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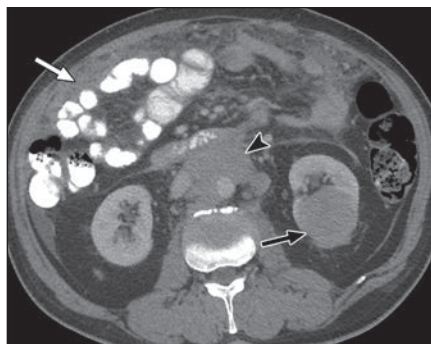
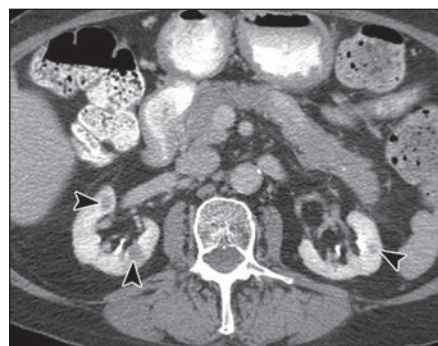
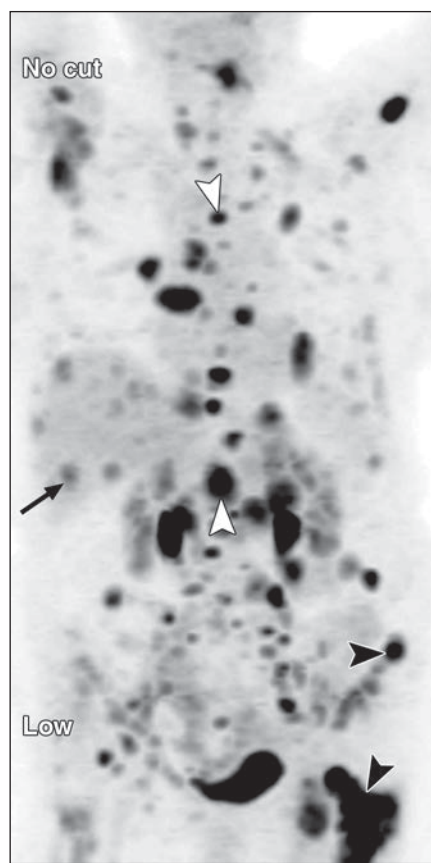


Fig. 3—53-year-old woman with large B-cell lymphoma. Axial nephrographic phase CT image shows 4-cm hypovascular mass (*black arrow*) in left kidney that enhances less than renal cortex. Retroperitoneal adenopathy (*arrowhead*) and peritoneal lymphomatosis (*white arrow*) are also seen.

Fig. 4—80-year-old woman with recurrent large B-cell lymphoma.
A, Axial nephrographic phase CT image shows multiple small (1–2-cm) bilateral hypovascular renal masses (*arrowheads*).
B, Coronal maximum-intensity-projection PET image shows widespread nodal (*white arrowheads*) and extranodal lymphoma involving bones (*black arrowheads*) and liver (*arrow*); hence, renal biopsy was not performed.



A



B

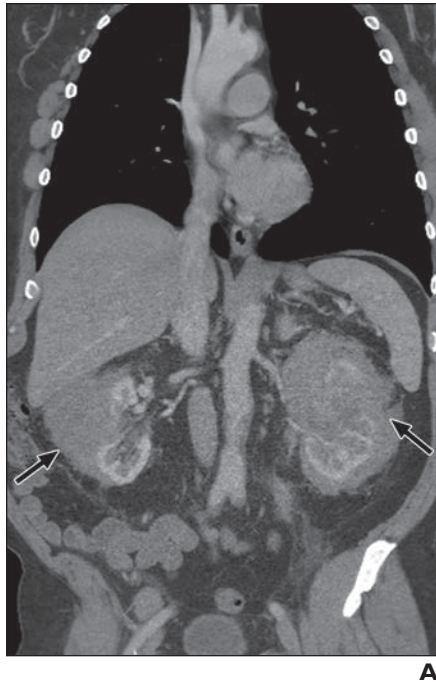


Fig. 5—64-year-old man with large B-cell lymphoma.

A, Coronal reformation image of corticomedullary phase CT shows bilateral perinephric soft-tissue masses (arrows). Tumor encases left kidney and extends into renal sinus. However, renal vessels are only pushed and are not invaded.

B, Coronal reformation image of nephrographic phase CT performed 3 months after chemotherapy shows almost complete resolution of perinephric masses.

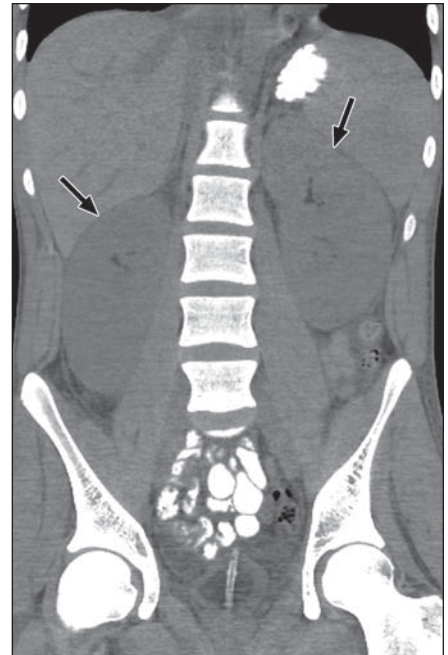
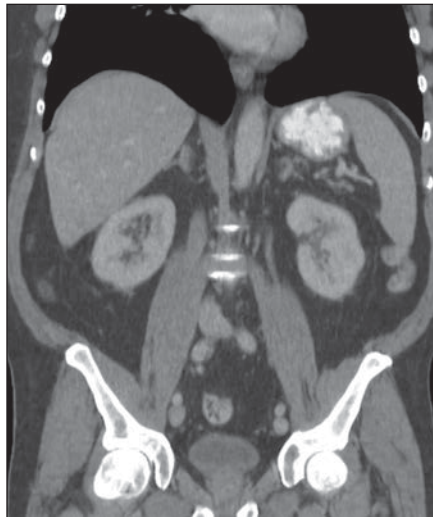


Fig. 6—48-year-old man who presented with acute renal failure. Coronal reformation image of unenhanced CT shows diffuse bilateral nephromegaly (arrows). Right kidney measured 14 cm and left kidney measured 15 cm. Percutaneous renal biopsy confirmed diagnosis of primary renal lymphoma.



Fig. 7—36-year-old man with low-grade follicular lymphoma. Axial CT image of abdomen shows soft-tissue mass (black arrowheads) within left renal sinus and renal pelvis. Enlarged retroperitoneal nodes (white arrowhead) are seen.



Fig. 8—76-year-old man with large B-cell lymphoma.

A, Axial CT image of abdomen shows 2.4-cm hypervascular mass (arrow) in interpolar region of right kidney. Enlarged retroperitoneal nodes (arrowhead) secondary to lymphoma are seen.

B, Axial CT image obtained in delayed phase shows washout of contrast material in renal mass (arrow). Hypervascular nature of mass makes renal lymphoma less likely. Considering patient's age and comorbid conditions and small size of renal mass, renal biopsy was performed.



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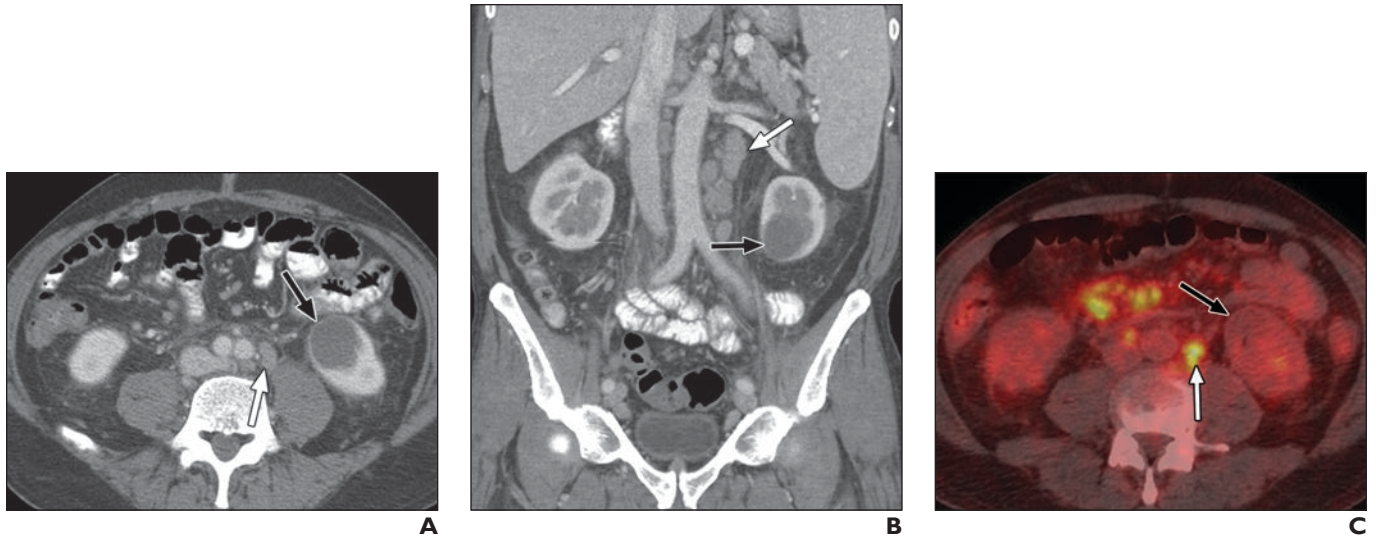


Fig. 9—60-year-old man with large B-cell lymphoma.

A, Axial CT image of abdomen shows 2.6-cm homogeneous hypovascular mass (*black arrow*) in lower pole of left kidney with average density of 38 HU. Retroperitoneal nodes (*white arrow*) are also present.

B, Coronal CT image of abdomen shows multiple enlarged retroperitoneal nodes (*white arrow*) and splenomegaly. *Black arrow* = mass.

C, Axial fused PET/CT image shows mass (*black arrow*) to be non-FDG-avid whereas nodes (*white arrow*) are FDG-avid. Although renal lymphoma is typically hypovascular, that this mass was not FDG-avid made renal lymphoma less likely. Percutaneous renal biopsy confirmed diagnosis of papillary renal cell carcinoma.